

Homework Set #3

5.19.1

$$\Omega^+ = \Omega \quad \therefore \exists S : S^+ S = 1 \quad \text{and}$$

$$S^+ \Omega S = \text{diag}(\omega_1, \dots, \omega_d) \equiv \Delta$$

$$S^+ f(\Omega) S = \sum_{n=0}^{\infty} S^+ \Omega^n S$$

$$= \sum_{n=0}^{\infty} (S^+ \Omega S)^n = \sum_{n=0}^{\infty} \Delta^n$$

$$= \sum_{n=0}^{\infty} \text{diag}(\omega_1^n, \dots, \omega_d^n)$$

$$= \text{diag}\left(\frac{1}{1-\omega_1}, \dots, \frac{1}{1-\omega_d}\right)$$

provided $|\omega_i| < 1$ all $i=1, \dots, n$

(otherwise the series does not converge)

$$= \frac{1}{1-\Delta} = \frac{1}{1-S^+ \Omega S}$$

$$\text{Thus } (1 - S^+ \Omega S) S^+ f(\Omega) S = 1$$

$$S^+ (1 - \Omega) f(\Omega) S = 1$$

$$(1 - \Omega) f(\Omega) = 1$$

$$\text{Similarly } f(\Omega) (1 - \Omega) = 1$$

S 1.9.2

$$U \equiv \sum_{n=0}^{\infty} \frac{1}{n!} i^n H^n$$

$$H^\dagger = H$$

$$U^\dagger = \sum_{n=0}^{\infty} \frac{1}{n!} (-i)^n H^n = e^{-iH}$$

$$U^\dagger U = e^{-iH} e^{iH} = \mathbb{1}$$

S 1.9.3

$$\exists S : S^\dagger H S = \text{diag}(\omega_1, \dots, \omega_d)$$

$$\equiv \Delta$$

$$S^\dagger U S = \sum_{n=0}^{\infty} \frac{1}{n!} i^n (S^\dagger H S)^n$$

$$= \text{diag}(e^{i\omega_1}, \dots, e^{i\omega_d})$$

$$\det(S^\dagger U S) = \det U$$

$$= e^{i(\omega_1 + \dots + \omega_d)} = e^{i \text{Tr} H}$$

S 1.10.1

$$\int_{-\infty}^{+\infty} f(x) \delta(ax) dx = \int_{-\infty}^{+\infty} f\left(\frac{y}{a}\right) \delta(y) \frac{dy}{a}$$

$$= \frac{1}{a} f(0) = \int_{-\infty}^{+\infty} f(x) \frac{1}{a} \delta(x) dx$$

for all $f(x)$

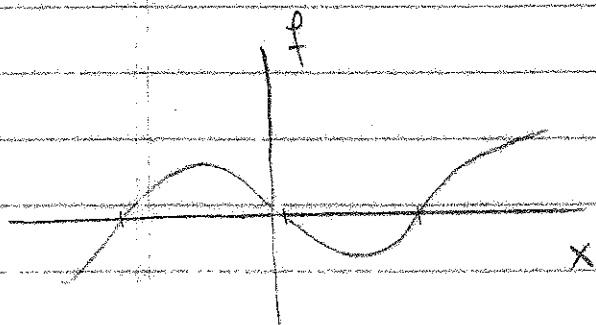
↖ not officially
part of the
homework

S 1.10.2

$$\int_{-\infty}^{+\infty} g(x) \delta[f(x)] dx$$

$$= \sum_i \int_{x_i - \epsilon}^{x_i + \epsilon} g(x) \delta[f(x)] dx$$

where the x_i are the
zeros of $f(x)$



We assume that the
 x_i are simple zeros.

Over the small interval
 $[x_i - \epsilon, x_i + \epsilon]$ we can
change variables $y_i = f(x)$

$$y_i = 0 + \frac{df}{dx}(x_i) (x - x_i) + \dots$$

$$x = x_i + \frac{1}{\frac{df}{dx}(x_i)} y_i + \dots$$

$$\int_{-\infty}^{+\infty} g(x) \delta[f(x)] dx$$

$$= \sum_i \int_{f(x_i - \epsilon)}^{f(x_i + \epsilon)} dy_i g\left(x_i + \frac{1}{\frac{df}{dx}(x_i)} y_i + \dots\right) \times$$

$$\times \delta(y_i) \frac{1}{\frac{df}{dx}(x_i) + \dots}$$

$$= \sum_i g(x_i) \frac{1}{\left| \frac{df}{dx}(x_i) \right|}$$

$$= \int_{-\infty}^{+\infty} dx g(x) \sum_i \delta(x - x_i) \frac{1}{\left| \frac{df}{dx}(x_i) \right|}$$

for all $g(x)$

S 1.10.3

$$\int_{-\infty}^x \delta(x') dx' = 1 \quad \text{if } x > 0$$

$$= 0 \quad \text{if } x < 0$$

definition of $\theta(x)$

$$\theta(x) = \int_{-\infty}^x dx' \delta(x')$$

this is not
part of the
homework

S 1.10.4

$$\epsilon \frac{\partial^2 \psi}{\partial t^2} - \tau \frac{\partial^2 \psi}{\partial x^2} = 0$$

ϵ = mass per unit length of the
string

τ = tension in the string

$$\psi(0) = \psi(L) = 0$$

$$\psi_n(x, t) = \sin\left(\frac{n\pi}{L} x\right) e^{-i\omega_n t}$$

with $\omega_n = \frac{n\pi}{L} \left(\frac{\tau}{\epsilon}\right)^{1/2}$

$\left(\frac{T}{\epsilon}\right)^{1/2}$ is the phase and group velocity of transverse oscillations of the string

$$\psi(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L} x\right) (A_n \cos \omega_n t + B_n \sin \omega_n t)$$

$$\dot{\psi}(x, 0) = 0 \quad \therefore B_n = 0$$

$$\psi(x, 0) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L} x\right)$$

$$\int_0^L dx \sin\left(\frac{m\pi}{L} x\right) \psi(x, 0)$$

$$= \int_0^L dx \sum_{n'=1}^{\infty} A_{n'} \sin\left(\frac{n'\pi}{L} x\right) \sin\left(\frac{m\pi}{L} x\right)$$

$$= \frac{L}{\pi} \sum_{n'=1}^{\infty} A_{n'} \int_0^{\pi} d\theta \underbrace{\sin n' \theta \sin n \theta}$$

$$\frac{1}{2} [\cos((n'-n)\theta) - \cos((n'+n)\theta)]$$

$$= \frac{L}{4\pi} \sum_{n'=1}^{\infty} A_{n'} \int_0^{2\pi} d\theta [\cos((n'-n)\theta) - \cos((n'+n)\theta)]$$

$$= \frac{L}{4\pi} \sum_{n'=1}^{\infty} A_{n'} 2\pi \delta_{nn'} = \frac{L}{2} A_n$$

$$= \int_0^{L/2} dx \frac{2xh}{L} \sin\left(\frac{n\pi x}{L}\right) + \int_{L/2}^L dx \frac{2h}{L} (L-x) \sin\left(\frac{n\pi x}{L}\right)$$

$x \rightarrow L-x$

$$= \int_0^{L/2} dx \frac{2xh}{L} \left[\sin\left(\frac{n\pi x}{L}\right) + \sin\left(n\pi - \frac{n\pi x}{L}\right) \right]$$

$-(-1)^n \sin\left(\frac{n\pi x}{L}\right)$

$$\left\{ \begin{array}{l} = 0 \quad \text{for } n \text{ even} \end{array} \right.$$

$$= \frac{4h}{L} \int_0^{L/2} dx x \sin\left(\frac{n\pi x}{L}\right) \quad \text{for } n \text{ odd}$$

$$= \frac{4h}{L} \left(\frac{L}{n\pi}\right)^2 \int_0^{n\pi/2} d\theta \theta \sin \theta$$

$$= \frac{4hL}{n^2 \pi^2} \left[-\theta \cos \theta + \sin \theta \right] \Big|_0^{n\pi/2}$$

$$= \frac{4hL}{n^2 \pi^2} \sin \frac{n\pi}{2}$$

$$A_n = \frac{8h}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$\psi(x,t) = \frac{8h}{\pi^2} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2}\right) \frac{1}{n^2} \sin\left(\frac{n\pi x}{L}\right) \cos \omega_n t$$